KINES 7103: Design and Analysis I

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HEALTH
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About Me…

- In my research, I want to improve our understanding of the mechanisms underlying neurological rehabilitation.
  - Ultimately I want to create more efficient, individualized rehabilitation approaches and improve the quality of life for individuals with disabilities.

- My background is in quantitative psychology, where I learned to model data and statistically test hypotheses about human learning.
  - I am accredited as a professional statistician ("Pstat"), by the *American Statistical Association*.
  - I have taught statistics in some capacity at either the undergraduate or graduate level since 2007.

What is KINES 7103?

- Design and Analysis I is a foundational research methods course for the College of Health. This means that…
  - We have students with a wealth of *different backgrounds* and *experiences*.
    - In class, I will try to find examples from the different disciplines in the CoH. In office hours, I can provide more personalized instruction and even work with you on your own data.
  - We need to introduce statistical techniques from across the *translational research pipeline*.
    - I have worked on computational models of neural processes, Phase III randomized controlled trials, and epidemiological data "in the wild".
  - In this class you will learn the complexities of statistical inference in the *general linear model*.
    - But remember that statistics is its own discipline. You wouldn't be an expert in genetics after taking 1-2 classes; you won't be an expert in statistics at the end of this class, and that's okay.

Introductory exercise…

- Scatter yourselves around the room in groups of 2-3. Try to sit with at least one person you don't already know.

- When I say "go", you have 30 seconds to draw your new partner. At the end of 30 seconds, you need to show your partner their portrait.

"The Scream"

[Edvard Munch, 1893]

"Creative Confidence"

[Bob McKim, Stanford University]

General Format for the Course

- We have a flipped classroom model. Every week, you will*:
  1. Read a chapter from the textbook.
  2. Watch (or re-watch) the pre-recorded video lectures.
  3. Complete an **Online Quiz** over the chapter.
  4. Attend class, in which I will lead you through some problems, answer your questions, and then you will work on the remaining problems to complete the **Lab Assignment**.
  5. Attend my office hours. You can follow-up with any questions you have or even bring in your own data that we can work on together.
     - I am genuinely happy for you to come to my office hours every week. You can treat it like a second class.
  6. Be sure to submit your lab assignment before the next class.

[* This is just my recommended order. You may find a different order works better.]

General Information

- We will teach statistics through the *general linear model*.
  - If you have had a statistics class before, you might have learned **a t-test** as a different test from **an ANOVA** as a different test from **an ANCOVA**, etc.
  - However, these are all forms of regression models.
    - Models that have a similar structure are given a certain name.
    - Rather than memorizing different families of test and computational short-cuts, we will learn a model building approach. The **GLM** is the most flexible approach and helps you properly understand the basis of tests.
- We will use algebra and geometry often.
  - Calculus will be helpful for understanding certain concepts, but is not essential to calculations or a broader understanding of the application.
    - $y_i = mx_i + b$
    - $y_i = \beta_0 + \beta_1(x_i)$
  - If you want proofs or more theoretical details for certain concepts, please come to my office hours.

General Information

- When you encounter equations in the text, you want to be able to read them like a sentence and ***understand the concepts***. For instance:

$$s_x = \sqrt{\frac{\sum(x_i - \bar{x})^2}{(n-1)}}$$

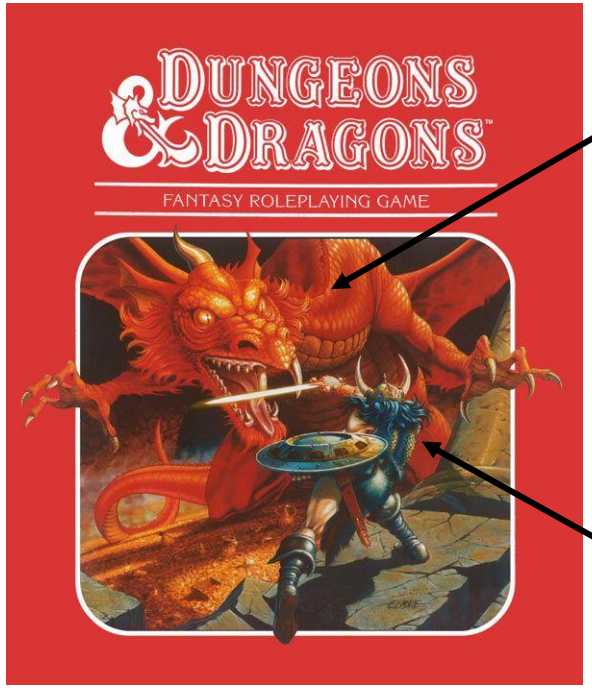
The standard deviation tells us how different values are on average. Specifically, how much a variable varies around its mean, standardized to the number of observations (more specifically the degrees of freedom).

- If you are having trouble understanding what an equation is saying conceptually:
  - Trying fixing all but one of the values. What happens to the outcome when you make that value bigger or smaller?
  - Shoot me an email,
  - Ask me in class,
  - Come to my office hours.

Gaining "Experience Points"

<u>In order of difficulty:</u>

1. **Online Quizzes**
2. **Lab Assignments**
3. **Quests**
   - There are two quests throughout the course. These are assigned and completed online.
   - Quests must be completed individually.
   - Copies of old quests are included in the compressed files for the course.



The unknown as personified by a dragon.

You.

[Larry Elmore, 1983]

Today

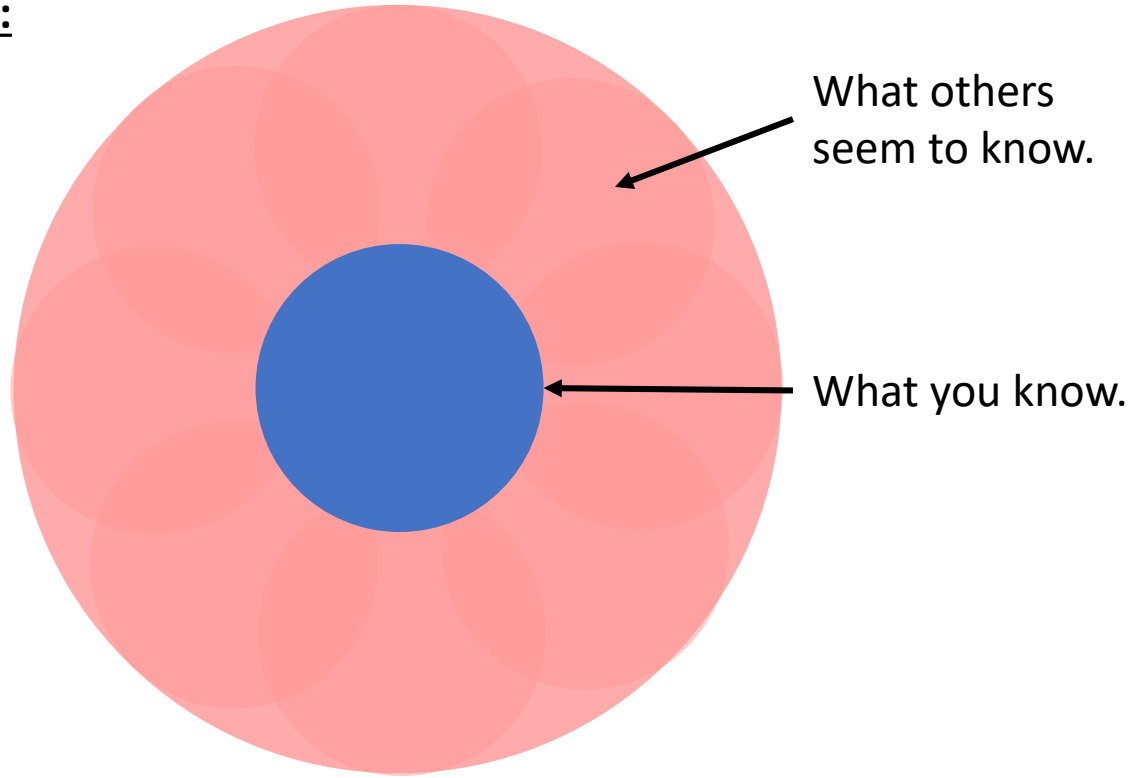
- Introduction to R and the R programming environment.
  - You do not need to know how to program in R to succeed in this course.
  - All* code will be provided for you.
  - R is free, powerful, and a valuable skill for both academia and industry.

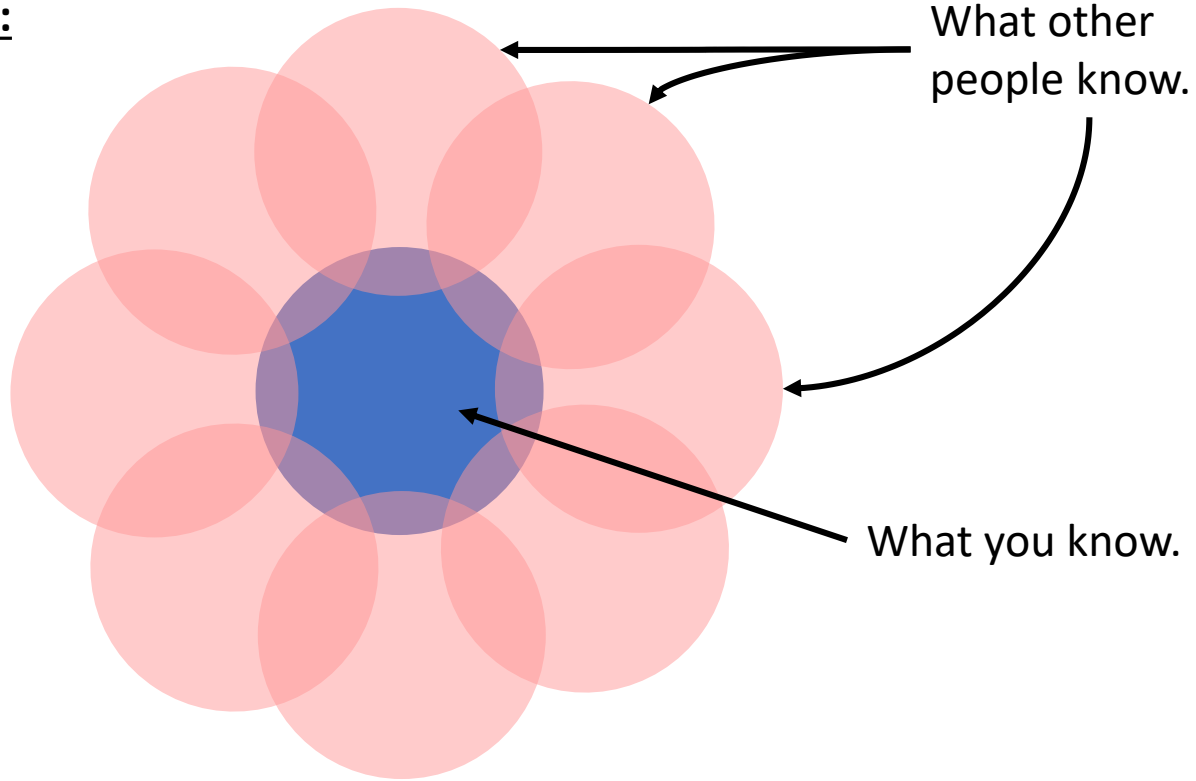
- Complete **Lab Assignment 01** and submit it through Canvas.
  - Please also be sure to watch Video 01 and complete **Quiz 01** by Wednesday if you have not already.

[*You may need to lightly edit code for lab assignments.]

Creative Confidence and Peer Judgment

Perception:



Creative Confidence and Peer Judgment

Perception: